

Topic 1.12 - Transformations of Functions

Practice Worksheet

Strengthened ECHS edition - reasoning, challenge, and AP-style practice

Name: _____

Class: _____

Date: _____

Directions

Show a mathematical argument, not only a final answer. Use exact values unless an approximation is requested. Calculator use is identified on each item. In context, state units, domain restrictions, and assumptions when relevant.

Learning focus

- Interpret and construct transformations in the form $g(x) = af(b(x-h)) + k$.
- Map points and key features between parent and transformed functions.
- Determine transformed domains, ranges, intercepts, asymptotes, and holes with justification.

Section A: Foundation and representation

Question 1

Interpretation

No calculator

Describe every transformation from $f(x)$ to $g(x) = -4f(2(x+3)) + 1$.

Question 2

Point mapping

No calculator

If $(8, -3)$ lies on f , find the corresponding point on $g(x) = 5f((x-2)/4) + 7$.

Section A continued

Question 3

Parameter

No calculator

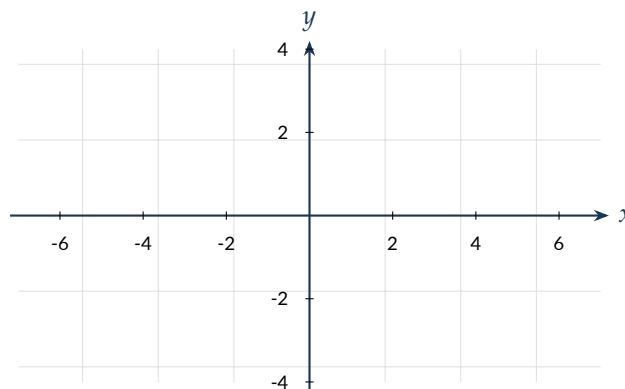
A point $(1, 6)$ on f maps to $(-2, -5)$ under $g(x) = af(3(x + 7/3)) + k$. If $a = -1$, find k .

Question 4

Quadratic

No calculator

Analyze and sketch $p(x) = \frac{1}{2}(x - 4)^2 - 3$: state transformations, vertex, axis, domain, range, and intercept strategy.



Question 5

Absolute value

No calculator

Analyze $v(x) = -2|x + 1| + 6$ and solve $v(x) = 0$.

Section B: Application and reasoning**Question 6**

Root function

No calculator

For $c(x) = \sqrt[3]{-2(x-5)} + 4$, identify the image of the parent origin and state domain and range.

Question 7

Rational

No calculator

Identify all asymptotes, domain, range, and intercepts of $r(x) = \frac{6}{x+2} - 1$.

Question 8

Restrictions

No calculator

The graph of f has a hole at $(2, -1)$. Find the transformed hole for $g(x) = 3f(-2(x-4)) + 5$.

Section C: Challenge and error analysis**Question 9**

Error analysis

No calculator

A student says $f(3(x - 2))$ means shift right 6 because the inside expands to $3x - 6$. Diagnose the reasoning.

Question 10

Construction

No calculator

Construct a transformed reciprocal function with vertical asymptote $x = 5$, horizontal asymptote $y = -2$, and passing through $(6, 1)$.

Question 11

Parameter

No calculator

Find a and h if $g(x) = a(x - h)^2 + 7$ has vertex $(3, 7)$ and passes through $(5, -1)$.

Question 12

Challenge

No calculator

Given $g(x) = -3f(2x - 8) + 5$, solve for $f(u)$ in terms of g and u .

Section D: AP-style multiple choice

MCQ 1

No calculator

If $(4, 7)$ lies on f , which point lies on $g(x) = 2f(x - 3) - 5$?

- A. $(1, 9)$
- B. $(7, 9)$
- C. $(7, 2)$
- D. $(1, 2)$

MCQ 2

No calculator

The graph of $f(5x)$ is a horizontal

- A. stretch by 5
- B. compression by $1/5$
- C. shift right 5
- D. vertical stretch by 5

MCQ 3

No calculator

What is the horizontal asymptote of $-4/(x + 7) + 3$?

- A. $x = -7$
- B. $x = 3$
- C. $y = -7$
- D. $y = 3$

MCQ 4

No calculator

If the range of f is $[1, 6]$, what is the range of $g(x) = -2f(x) + 5$?

- A. $[-7, 3]$
- B. $[3, 17]$
- C. $[-12, -2]$
- D. $[-17, -3]$

AP-style free response 1

No calculator - 6 points

The table gives selected points on f .

x	-4	-1	2	5
$f(x)$	3	-2	5	1

Define $g(x) = -2f(3(x - 1)) + 4$.

A. Create the corresponding table of four points on g .

B. If the domain of f is $[-4, 5]$, find the domain of g .

C. If the range of f is $[-2, 5]$, find the range of g .

D. Explain how the sign of the outside scale affects the order of output values.

AP-style free response 2**Calculator permitted - 6 points**

A transformed reciprocal model has the form $R(x) = a/(x - h) + k$. Its vertical asymptote is $x = 4$, horizontal asymptote is $y = -2$, and the graph passes through $(6, 3)$.

A. Determine a, h, k and write the model.

B. Find both intercepts, giving decimal approximations if needed.

C. Describe the one-sided behavior at the vertical asymptote.

D. State domain and range and explain why neither asymptote is part of the graph.
